

Question	Marking Guidance	Mark	Comments
05.1	1. Where dividing cells are found / mitosis occurs; OR No dividing cells / mitosis in tissue further away / more than 5mm from tip; OR To get (soft) tissue that will squash; OR Length that will fit under cover slip; 2. Single / thin layer of cells / spread out cells so light passes through (making cells / nuclei visible);	2	1. Accept most dividing cells 2. Accept thin layer of tissue 2. Ignore to see cells clearly
05.2	3.57 / 3.6 / 3.7 / 3.71 / 3.8 (%);;	2	If the answer includes additional decimal places, award the marks if it would round to a correct answer There are 3 cells in anaphase Accept for 1 mark, 101.25 / 101 (students estimate in minutes) OR 3.75 (difference between scientist estimate and student's estimate in minutes) Ignore plus or minus signs
05.3	Cytokinesis;	1	

05.4	Description; Explanation; E.g, 1. Examine large number of fields of view / many cells; 2. To ensure representative sample; OR 3. Repeat count; 4. To ensure figures are correct; OR 5. Method to deal with part cells shown at edge /count only whole cells; 6. To standardise counting;	2 max	Mark as pairs only 1. Accept large number / 20 or more for many 2. Accept typical / reliable
05.5	1. Stops anaphase / cell division / mitosis; 2. (By) stopping / disrupting / spindle fibres forming / attaching / pulling; 3. Preventing separation of (sister) chromatids; 4. (So) no new cells added (to root tip);	3 max	1. Accept prevents telophase / cytokinesis 2. Ignore affects anaphase 3. Ignore chromosomes separate / split 3. Accept chromatids split